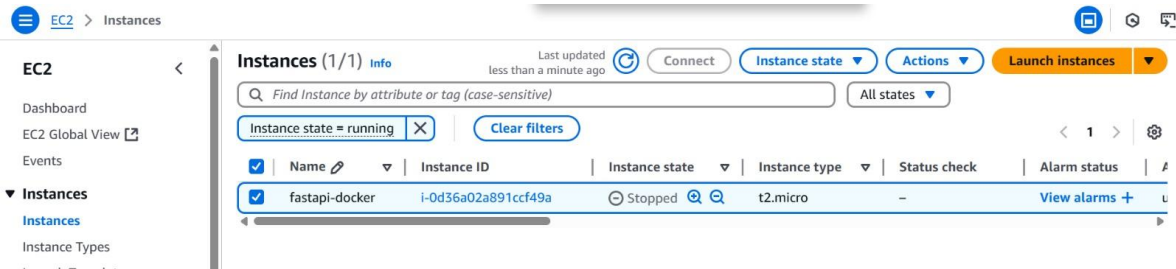


FastAPI in AWS

1. Generate the instance in AWS



2. Clone the github repository

```
ubuntu@ip-172-31-21-196:~$ git clone https://github.com/McGill-MMA-EnterpriseAnalytics/Superstore-Marketing-Campaign.git
Cloning into 'Superstore-Marketing-Campaign'...
remote: Enumerating objects: 2724, done.
remote: Counting objects: 100% (256/256), done.
remote: Compressing objects: 100% (112/112), done.
remote: Total 2724 (delta 102), reused 196 (delta 65), pack-reused 2468 (from 1)
Receiving objects: 100% (2724/2724), 32.84 MiB | 2.08 MiB/s, done.
Resolving deltas: 100% (978/978), done.
ubuntu@ip-172-31-21-196:~$ cd Superstore-Marketing-Campaign
```

3. Build and run the image from the docker

```
ubuntu@ip-172-31-21-196:~/Superstore-Marketing-Campaign$ docker build -t superstore-api -f dockerfile.api .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 47.66MB
Step 1/11 : FROM python:3.10-slim AS builder
--> b32fa0454ca1
Step 2/11 : RUN apt-get update && apt-get install -y --no-install-recommends libgomp1 && rm -rf /var/lib/apt/lists/*
--> Using cache
--> f9fb1be054cf
Step 3/11 : WORKDIR /app
--> Using cache
--> abf9a4b659df
Step 4/11 : RUN pip install fastapi uvicorn pandas numpy scikit-learn joblib catboost
--> Using cache
--> a65a2c66abad
Step 5/11 : COPY models/best_model_tuned.pkl /app/models/
--> 5861665a991d
Step 6/11 : COPY src/app.py /app/src/
--> 4bd56c800eda
Step 7/11 : COPY src/utils/ /app/src/
--> 71d82661bdc7
Step 8/11 : COPY config.yaml /app/config.yaml
--> 7aff5d77cc6d
Step 9/11 : ENV PYTHONPATH=/app
--> Running in 9f605e3f7fcd
--> Removed intermediate container 9f605e3f7fcd
--> 4972614705f3
Step 10/11 : EXPOSE 8000
--> Running in c3e564f482b5
--> Removed intermediate container c3e564f482b5
--> c2ab400b271d
Step 11/11 : CMD ["python", "-m", "uvicorn", "src.app:app", "--host", "0.0.0.0", "--port", "8000"]
--> Running in 74c2c73795cf
--> Removed intermediate container 74c2c73795cf
--> fd2172efcd5c
Successfully built fd2172efcd5c
Successfully tagged superstore-api:latest
```

```
ubuntu@ip-172-31-21-196:~/Superstore-Marketing-Campaign$ docker run -d -p 8000:8000 superstore-api
86d684bb167946ced8ccddb6082008950ede12c354eecac663cad8daea43bb2
```

4. Ensure that the fastAPI is running

```
Create and run a new container from an image
ubuntu@ip-172-31-21-196:~/Superstore-Marketing-Campaign$ docker images
REPOSITORY      TAG       IMAGE ID       CREATED        SIZE
superstore-api   latest    fd2172efcd5c   About a minute ago   1.16GB
python          3.10-slim b32fa0454ca1   3 weeks ago       127MB
hello-world      latest    74cc54e27dc4   3 months ago       10.1kB
ubuntu@ip-172-31-21-196:~/Superstore-Marketing-Campaign$ docker run -d -p 8000:8000 superstore-api
86d684bb167946ced8ccddb6082008950ede12c354eecac663cad8daea43bb2
ubuntu@ip-172-31-21-196:~/Superstore-Marketing-Campaign$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
86d684bb1679   superstore-api "python -m uvicorn s..." 32 seconds ago Up 30 seconds 0.0.0.0:8000->8000/tcp, :::8000->8000/tcp
zen_volhard
```

5. Public terminal running in the cloud (<http://3.129.62.135:8000/docs>)

Superstore Marketing Campaign Prediction API 1.0.0 OAS 3.1

/openapi.json

API for predicting marketing campaign response

default

- GET** / Read Root
- GET** /health Health Check
- POST** /predict Predict
- POST** /predict-batch Predict Batch

Schemas

- HTTPValidationError** > Expand all object
- PredictionInput** > Expand all object